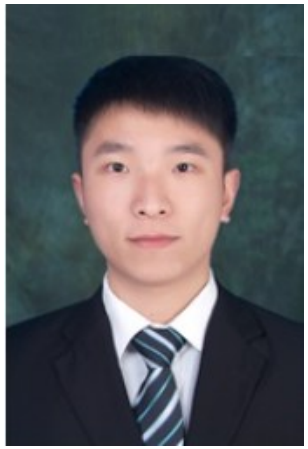
LSGI3245 – Geospatial Database Management and Design

Lecture 1 - Introduction

Instructor: Yue YU
Department of Land Surveying and Geospatial Science,
The Hong Kong Polytechnic University

Teaching Staff



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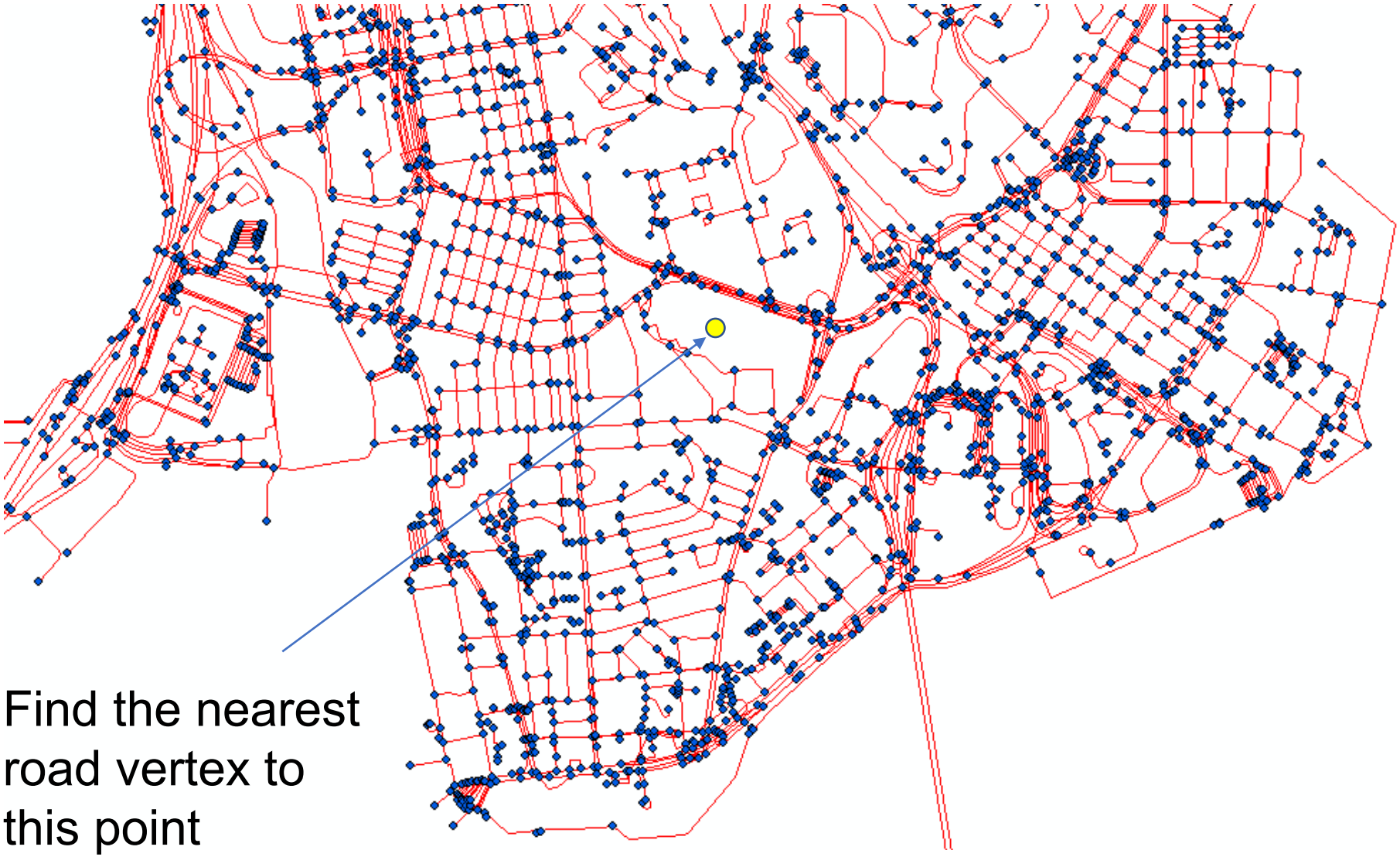
Questions

- Have you studied/used database systems before?



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Source: Google Image



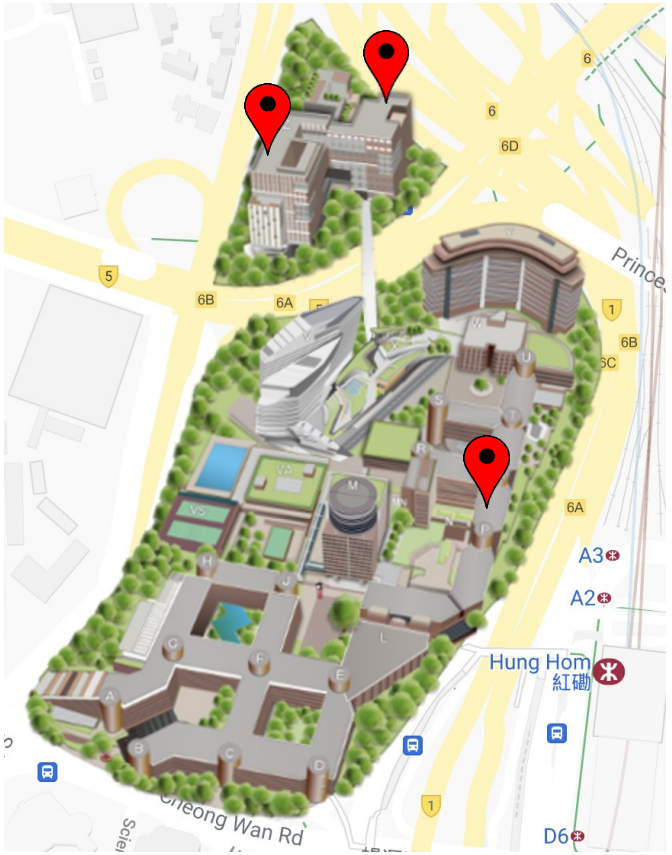
Find the nearest road vertex to this point

Course Information

Course Website: Learn@PolyU

Course Schedule

- 13 teaching weeks (lectures + labs)
- Lectures (Tuesday)
  - Time: 08:30 – 10:20
  - Venue: **FJ304**
- Labs (Wednesday)
  - Time: 16:30 – 18:20
  - Venue: **ZN604**, **ZS1001**
- Number of students: 118



Source: PolyU

Learning Objectives



- Understand the concepts of geospatial database, data models, and DBMS
- Implement and use a relational database
- Implement and use a geospatial database
- Use geospatial database and relevant techniques to tackle different GIS tasks

Source: Google Image

Assessment

Continuous Assessment + Examination

- Phase Test 1 (20%)
- Phase Test 2 (15%)
- Project (15%)
- Final Examination (50%)



* Closed-book exam

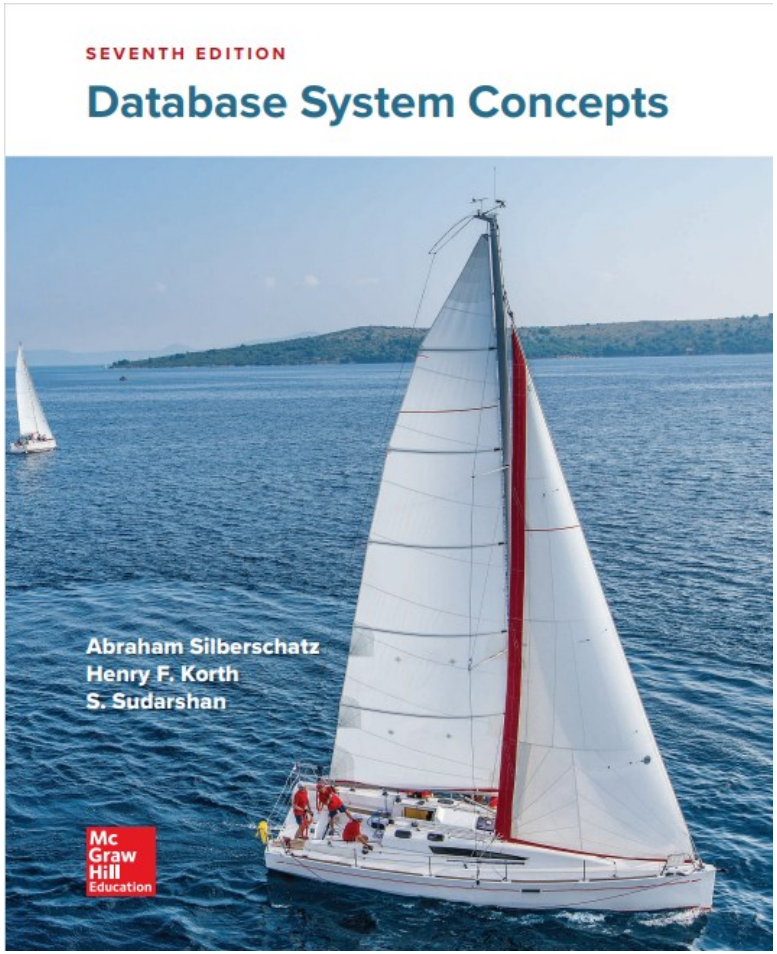
Reading List and References

Recommended:

1. Obe, R. O., & Hsu, L. S. (2021).PostGIS in action (3rd ed.). Manning Publications.
2. Mamoulis, N. (2012). Spatial data management. Morgan & Claypool Publishers.
3. Wu, Q. (2025). Spatial data management with DuckDB: From SQL basics to advanced geospatial analytics. Leanpub.
4. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2024). Database system concepts (8th ed.). McGraw-Hill.
5. Chang, K. T. (2019). Introduction to geographic information systems (9th ed.). McGraw-Hill.
6. M. F. Worboys and M. Duckham, "*GIS, a Computer Perspective*", online resources available at http://worboys.duckham.org

Supplementary:

1. A. K. W. Yeung and G. B. Hall, "*Spatial Database Systems – Design, Implementation and Project Management*", Springer, 2007, PolyU call number G70.212.Y38 2007eb (electronic resource)
2. PostgreSQL documentation website http://www.postgresql.org/docs
3. PostGIS documentation website http://postgis.net/documentation



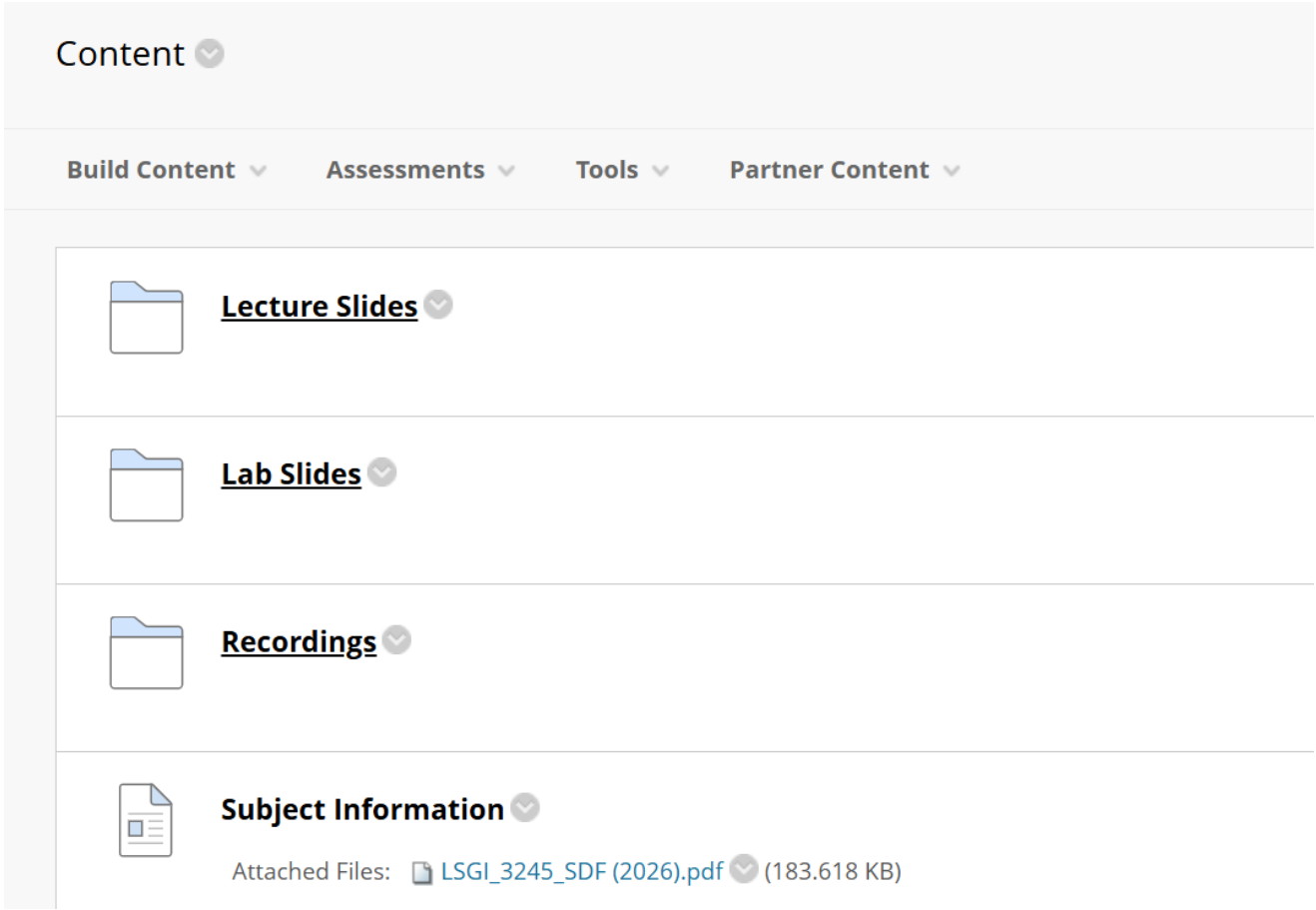
Source: Google Image

Tentative Schedule

Week	Lecture (Tuesday)		Lab (Wednesday)	
1	1-Sept	Course Introduction	Lab Introduction and Software Installation	2-Sept
2	8-Sept	Relational Model	Introduction to PostgreSQL (Part 1)	9-Sept
3	15-Sept	SQL (Part 1)	Introduction to PostgreSQL (Part 2)	16-Sept
4	22-Sept	SQL (Part 2) Release Information for Final Project	Query and Data Manipulation Using SQL (Part 1)	23-Sept
5	29-Sept	Intermediate SQL	Query and Data Manipulation Using SQL (Part 2)	30-Sept
6	6-Oct	Database Design	Introduction to PostGIS	7-Oct
7	13-Oct	Spatial Database & PostGIS	PostGIS: Spatial Datatype and Relationships	14-Oct
8	20-Oct	Test 1	PostGIS: Spatial Queries	21-Oct
9	27-Oct	PostGIS & Spatial Queries	Students Work on Final Projects (Part 1)	28-Oct
10	3-Nov	Spatial Storage and Indexing	Students Work on Final Projects (Part 2)	4-Nov
11	10-Nov	Test 2	Students Compile Final Project Reports (Part 1)	11-Nov
12	17-Nov	Project Presentation (Part 1)	Students Compile Final Project Reports (Part 2)	18-Nov
13	24-Nov	Project Presentation (Part 2)	Summary and Outlook	25-Nov

More Course Information

- Course Website: Learn@PolyU



Let's get started!

Content

- Database Management System (DBMS)
- Data Abstraction
- Database Languages
- Database Design
- Data Storage and Querying
- Database Users and Administrators

Learning Outcomes of Today's Class



- What a database management system (DBMS) is?
- The three levels of abstraction provided by the DBMS
- The database languages (i.e., how to communicate with the database system to achieve certain tasks?)
- Concept of relational database
- Who uses database systems?

Database Management System

Database Management System (DBMS)

- A database management system (DBMS) is a collection of interrelated data and a set of programs to access those data.
- The primary goal of a DBMS is to provide a way to store and retrieve database information that is both convenient and efficient.
- DBMS is used to manage collections of data that are:
  - Highly valuable
  - Relative large
  - Accessed by multiple users and applications (often at the same time)



Source: Google Image

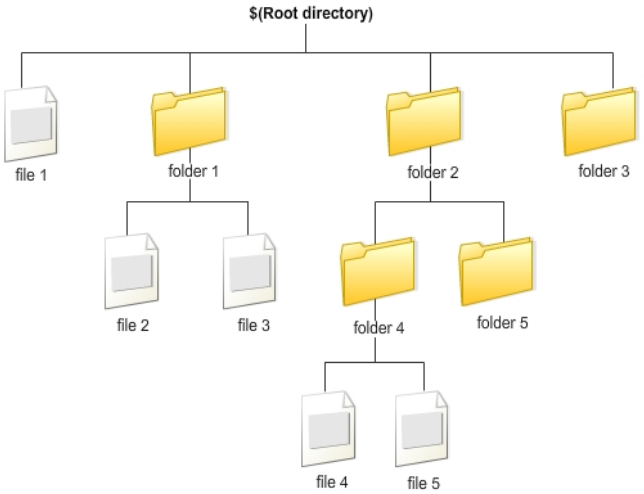
Database Applications

- Enterprise Information
  - Sales: customers, products, purchases
  - Human Resources: employees, salaries, payroll taxes
- Banking
  - Customer information, accounts, loans, and banking transactions
- Universities
  - Students, staff, courses, registration
- Web-based Services:
  - Social media: users, connections between users, posts, rating

Databases almost touch all aspects of our lives!

Manage Data Using File Systems

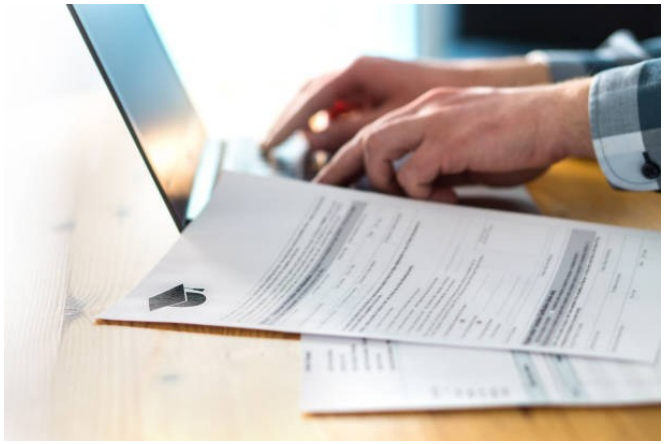
- In the early days, database applications were built directly on top of file systems.
- The system stores permanent records in various files, and it needs different application programs to extract records from, and add records to, the appropriate files.
- New application programs are added to the system as the need arises.



Source: Google Image

Manage Data Using File Systems (Cont.)

- University database example
  - Data: students, instructors, courses
- Application programs
  - Add new students, instructors, and courses
  - Register students for courses, and generate class rosters
  - Assign grades to students, compute grade point averages (GPA) and generate transcripts



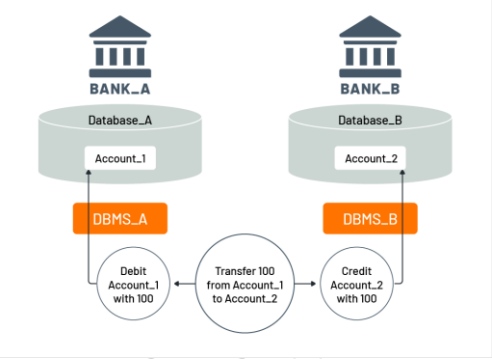
Source: Google Image

Drawbacks of Using File Systems

- Data redundancy and inconsistency
  - Multiple file formats, duplication of information in different files
- Difficulty in accessing data
  - Need to write a new program to carry out each new task
- Data isolation
  - Multiple files and format

Drawbacks of Using File Systems (Cont.)

- Integrity problems
  - The data values stored in the database must satisfy certain types of consistency constraints.
- Atomicity of updates
  - Failures may leave the database in an inconsistent state with partial updates carried out (**Example**: Transfer of funds from one account to another should either complete or not happen at all)



Source: Google Image

Drawbacks of Using File Systems (Cont.)

- Concurrent access by multiple users
  - Uncontrolled concurrent accesses can lead to inconsistencies (**Example**: Two people reading a balance and updating it by withdrawing money at the same time)

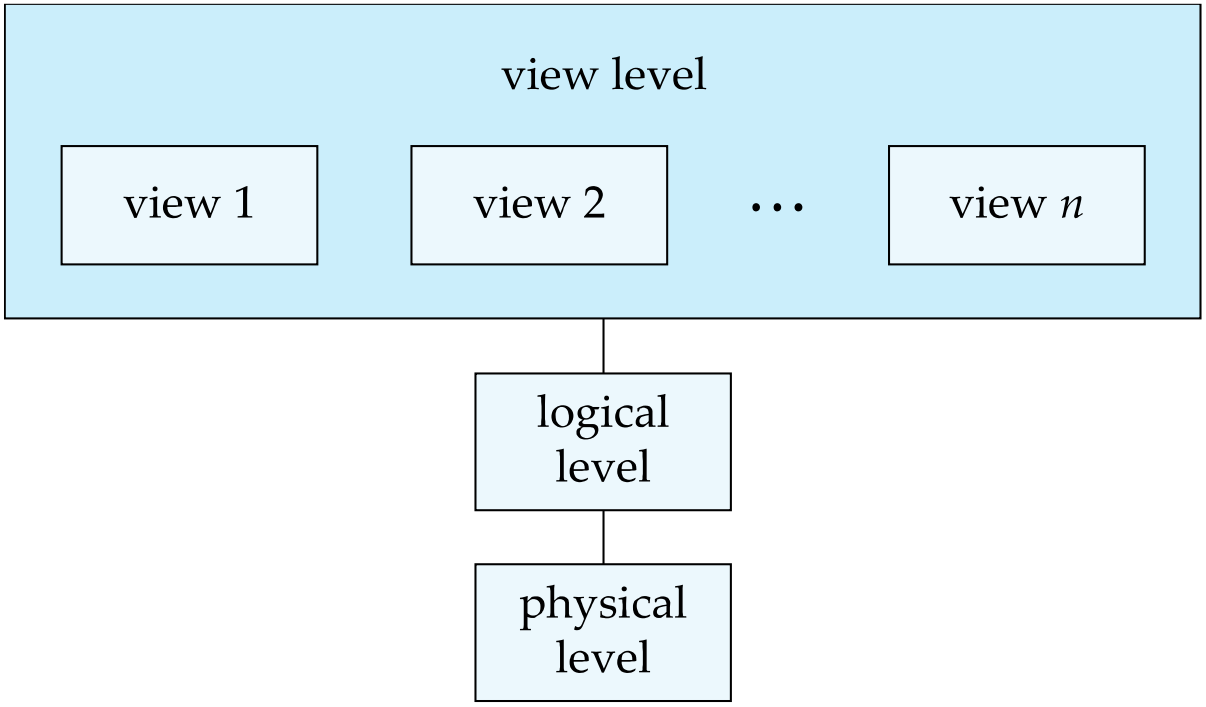
- Security problems
  - Hard to provide user access to some, but not all, data

DBMS offers solutions to all the above problems!

Data Abstraction

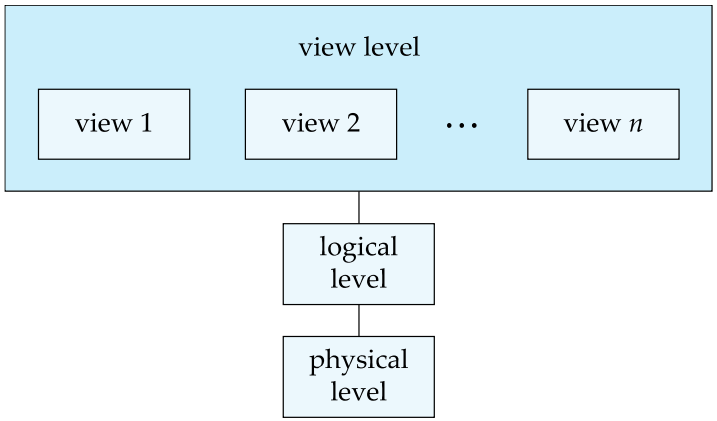
Levels of Abstraction

A major purpose of a database system is to provide users with an abstract view of the data, hiding complexity from users.



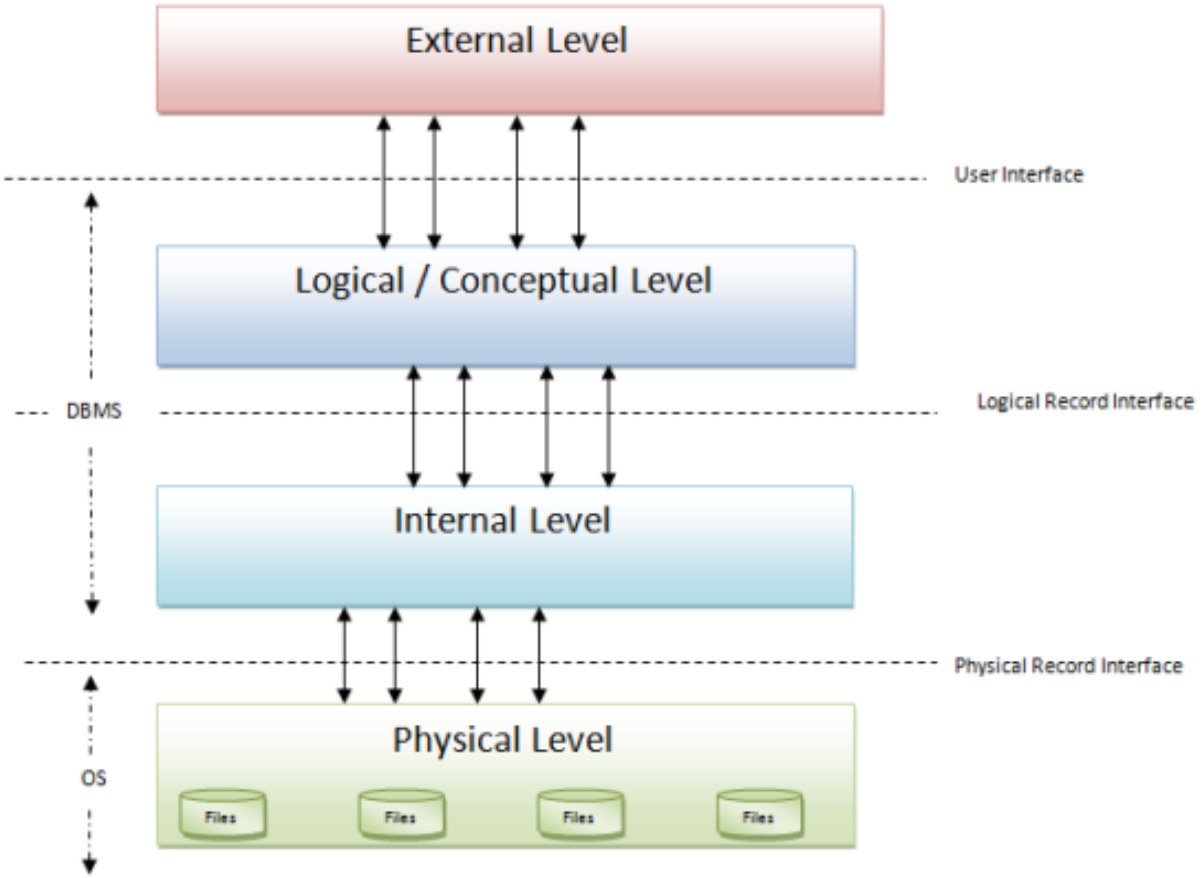
Levels of Abstraction (Cont.)

- Physical level: describes how data are stored. (lowest level)
- Logical level: describes what data are stored in database, and the relationships among the data.
- View level: describes only part of the entire database. Views can hide information for simplicity or security purposes. (highest level)



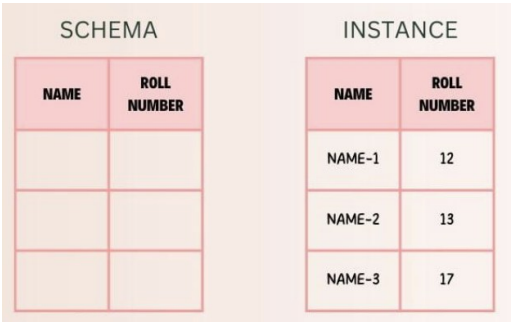

```
type instructor = record
                    ID : string;
                    name : string;
                    dept_name : string;
                    salary : integer;
                     end;
```

An Architecture of A database System



Source: https://www.tutorialcup.com/dbms/database-abstraction.htm

Instances and Schemas



- Instance
  - The actual content of the database at a particular point in time
  - Analogous to the value of a variable in programming languages
- Schema
  - The overall design of the database
    - Logical schema: The overall logical structure of the database
      - Example: The database consists of information about a set of customers and accounts in a bank and the relationship between them.
      - Analogous to type of a variable in in programming languages
    - Physical schema: The overall physical structure of the database

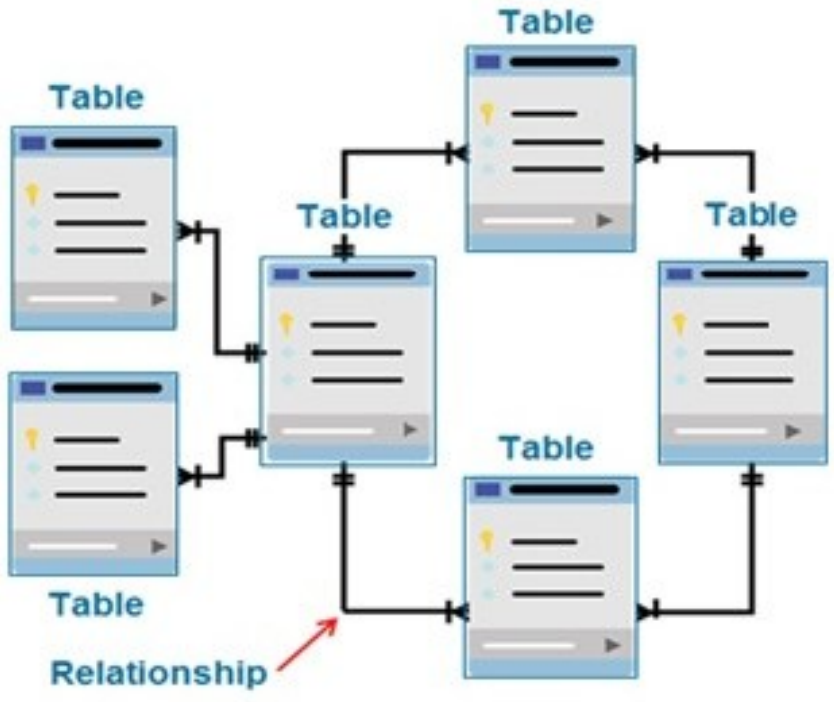
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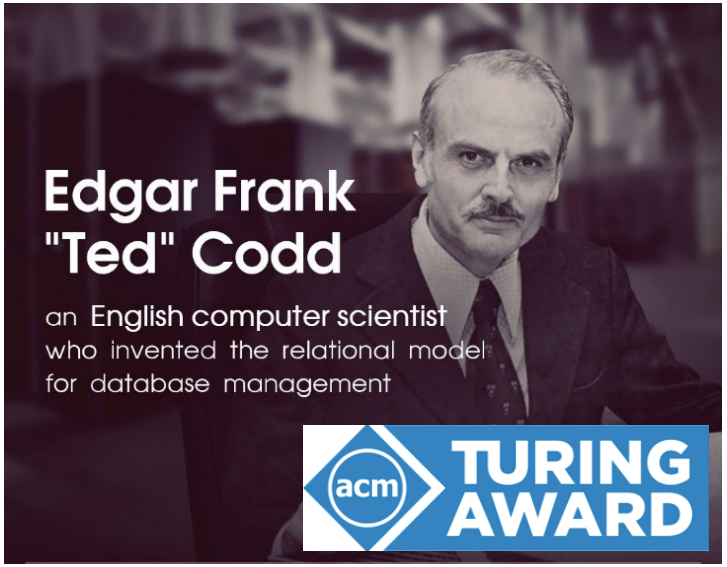
Data Models

- A collection of conceptual tools for describing data, data relationships, data semantics, and consistency constraints.
- Different categories of data models:
  - Relational Model (uses a collection of tables to represent data and relationships; most widely used for database today)
  - Entity-Relationship Model (uses a collection of basic objects/entities and their relationships, widely used in database design)
  - Object-Based Data Model (led by the trend of object-oriented programming such as in Java, C++; integrated into relational model today)
  - Semi-structured Data Model (permits data items of the same type to have different attributes, e.g., XML and JSON)

What is Relational Model?

- The relational model uses a collection of tables to represent both the data and the relationships among those data.

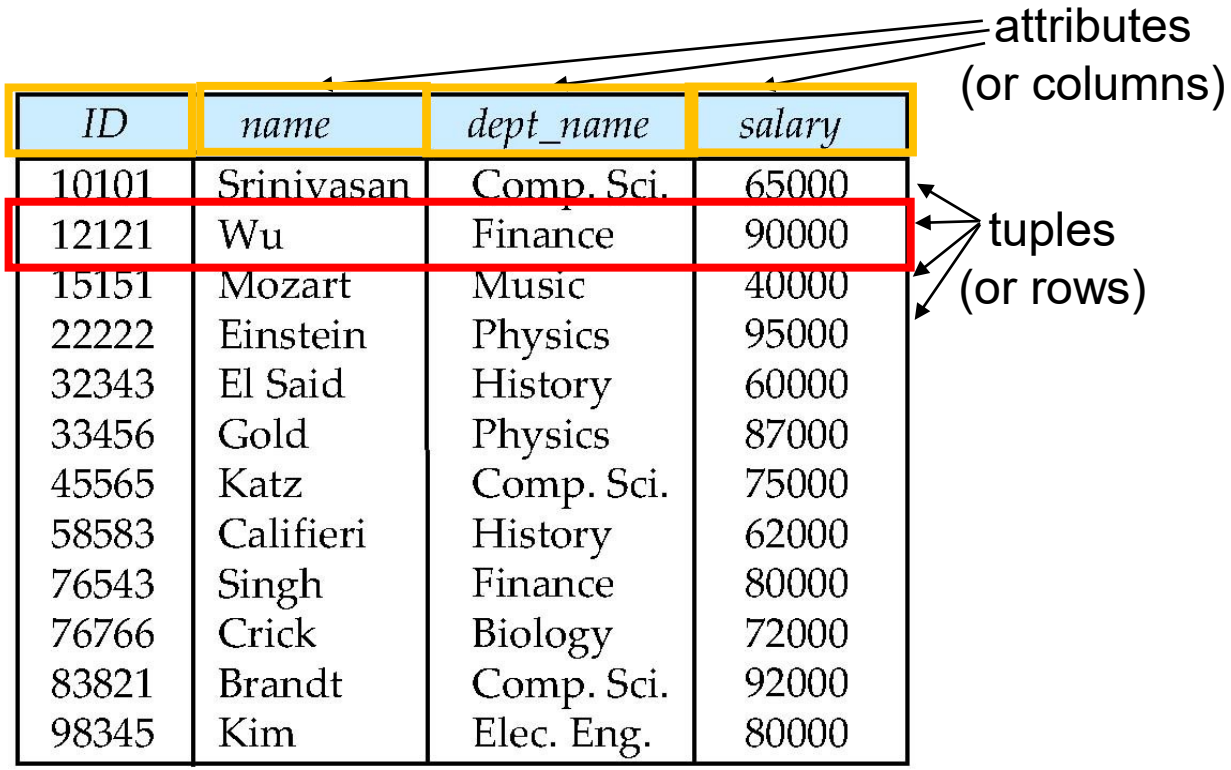




Source: Google Image

What is Relational Model?

- All the data is stored in various tables.
- Example of tabular data in the relational model



ID	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

Source: http://www.db-book.com/

A Sample Relational Database

ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

(a) The *instructor* table

dept_name	*building*	*budget*
Comp. Sci.	Taylor	100000
Biology	Watson	90000
Elec. Eng.	Taylor	85000
Music	Packard	80000
Finance	Painter	120000
History	Painter	50000
Physics	Watson	70000

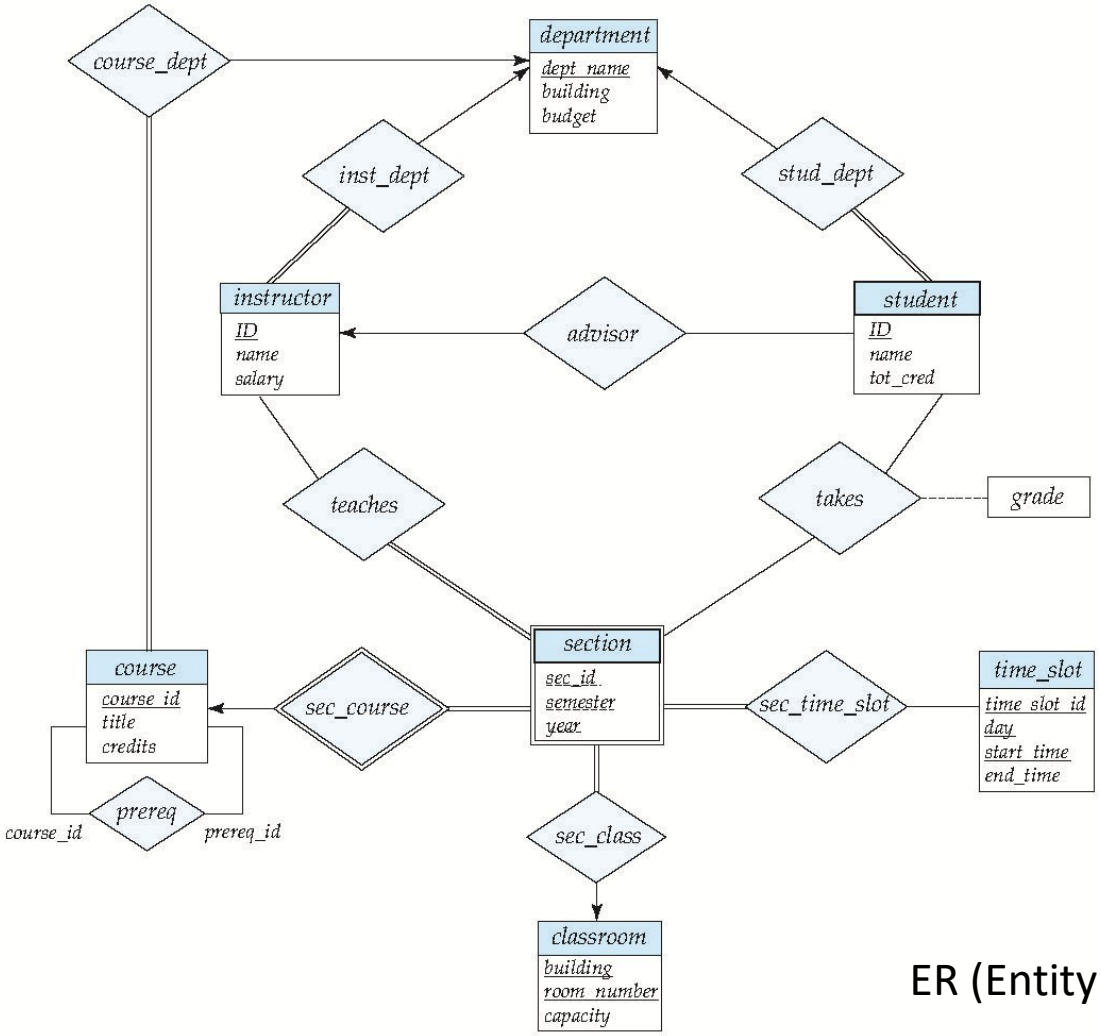
(b) The *department* table

A **relational database** consists of a collection of tables, each of which is assigned a unique name.

create table *instructor* (
ID **char**(5),
name **varchar**(20),
dept_name **varchar**(20),
salary **numeric**(8,2))

create table *department* (
dept_name **varchar**(20),
building **varchar**(20),
budget **numeric**(8,2))

ER Diagram for a University Enterprise



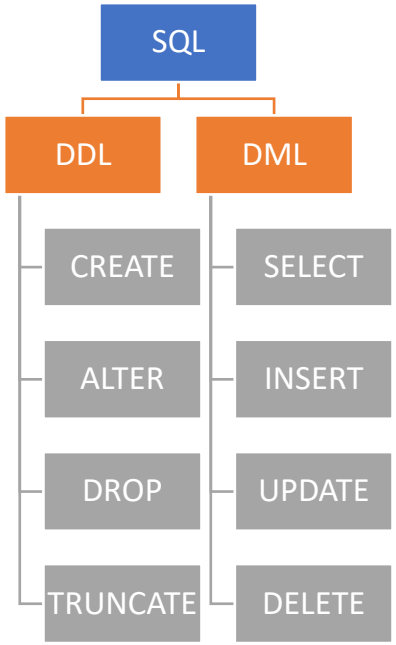
ER (Entity Relationship)

Database Languages

Database Languages

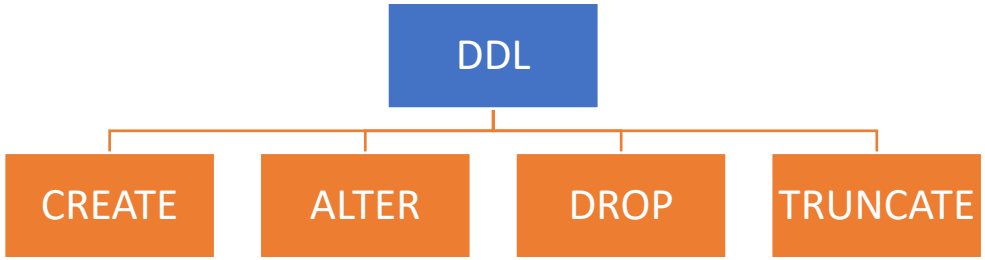
- Data-definition language (DDL)
  - To specify the database schema
- Data-manipulation language (DML)
  - To express database queries and updates

- In practice, they are not separate languages. They simply form parts of a single database language, such as the widely used SQL language.



Data-Definition Language (DDL)

- The DDL is used to define the database schema; specify the storage structure and access methods used by the database system.



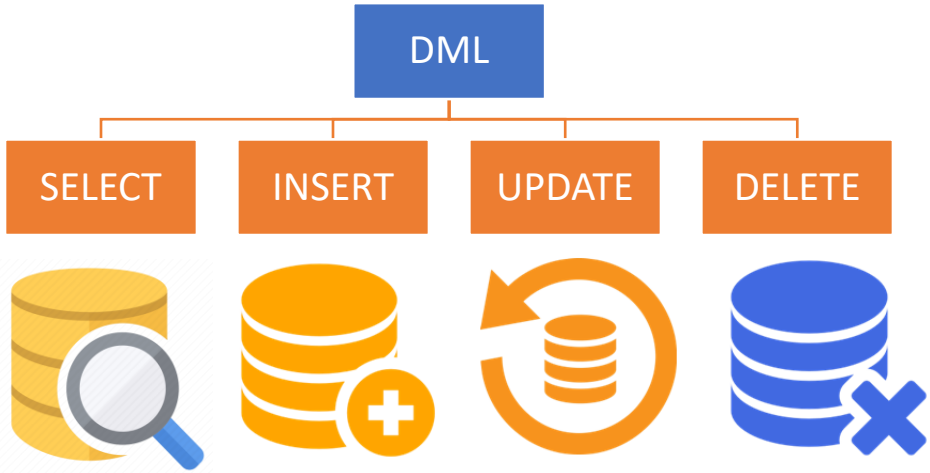
Example: **CREATE table** *instructor* (
ID **char**(5),
name **varchar**(20),
dept_name **varchar**(20),
salary **numeric**(8,2))

Data Definition Language (DDL)

- The data values stored in the database must satisfy certain consistency constraints.
- The DDL provides facilities to specify such constraints:
  - **Domain constraints**
    - The value of an attribute must follow certain data type, e.g., integer, character
  - **Referential integrity**
    - Ensure that the value that appears in one table, also appears in another table (Example: '*dept_name*' value in '*instructor*' table must appear in the '*dept_name*' attribute of '*department*' table)
  - **Authorization**
    - Who can access what; to differentiate among the users based on their rights to access the database

Data Manipulation Language (DML)

- The DML is a language that enables users to access or manipulate data as organized by the appropriate data model.
  - Retrieval of information stored in the database
  - Insertion of new information into the database
  - Modification of information stored in the database
  - Deletion of information from the database



SQL example:

```
SELECT name
FROM instructor
WHERE dept_name = 'LSGI'
```


Source: Google Image

Data Manipulation Language (DML)

- Two basic types of DML:
  - Procedural: need to specify what data are needed and how to get those data.
  - Declarative (non-procedural): specify what data are needed ***without*** specifying how to get those data. SQL is a non-procedural language.



Source: https://365datascience.com/

Database Design

Database Design

- Initial phase
  - characterize fully the data needs of the prospective database users
- Conceptual design
  - Applying the concepts of the chosen data model and translating user requirements into a conceptual schema of the database.
  - The designer reviews that schema to confirm that all data requirements are indeed satisfied and are not in conflict with one another.
- Logical and physical design
  - The process of moving from an abstract data model to the implementation of the database
  - Logical design: Deciding on the database schema
  - Physical design: Deciding on the physical layout of the database

Group Discussion



- Is there any problem with this design?

ID	*name*	*salary*	*dept_name*	*building*	*budget*
22222	Einstein	95000	Physics	Watson	70000
12121	Wu	90000	Finance	Painter	120000
32343	El Said	60000	History	Painter	50000
45565	Katz	75000	Comp. Sci.	Taylor	100000
98345	Kim	80000	Elec. Eng.	Taylor	85000
76766	Crick	72000	Biology	Watson	90000
10101	Srinivasan	65000	Comp. Sci.	Taylor	100000
58583	Califieri	62000	History	Painter	50000
83821	Brandt	92000	Comp. Sci	Taylor	100000
15151	Mozart	40000	Music	Packard	80000
33456	Gold	87000	Physics	Watson	70000
76543	Singh	80000	Finance	Painter	120000

Source: http://www.db-book.com/

Group Discussion



ID	name	salary	dept_name	building	budget
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(a) The *instructor* table

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Music	Packard	80000
Finance	Painter	120000
History	Painter	50000
Physics	Watson	70000

(b) The *department* table

Which one is better?

Issues About This Design

- Redundancy of information
  - The building and budget of departments
- Complication of update
  - Changing the budget of a department needs to be reflected in multiple rows
- Inability to represent certain information
  - Inserting a new department without information of faculty members (ID, name, salary).



ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
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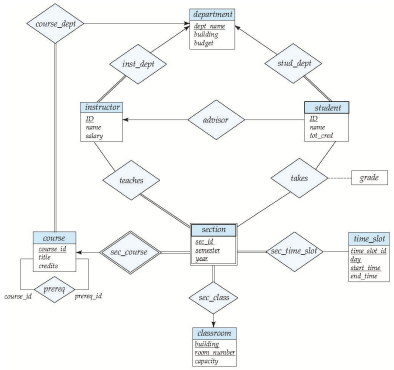
(b) The *department* table



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33456	Gold	87000	Physics	Watson	70000
76543	Singh	80000	Finance	Painter	120000

Database Approaches

- Need to come up with a methodology to ensure that each of the relations in the database is “good”.
- Two ways of doing so:
  - Entity relationship (E-R) data model: models an enterprise as a collection of entities and relationships.



  - Normalization theory: the goal is to generate a set of relation schemas that allows us to store information without unnecessary redundancy, yet also allows us to retrieve information easily.

Source: http://www.db-book.com/

Data Storage and Querying

Database Engine

- A database system is partitioned into modules that deals with different functions of the overall system

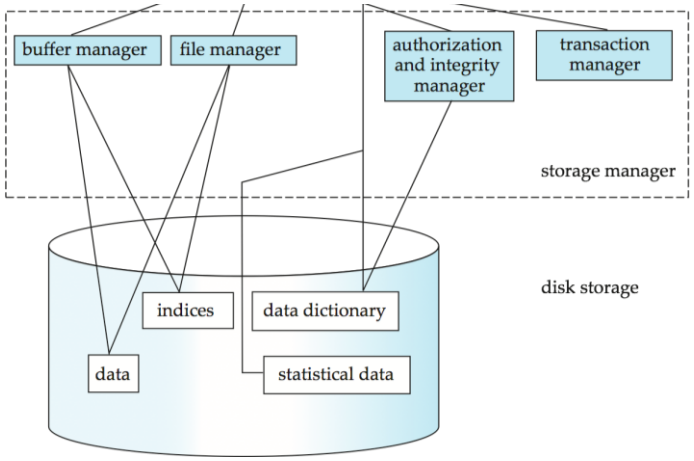


- The functional components can be broadly divided into:
  - The storage manager
  - The query processor

Source: Google Image

Data Storage and Querying

- The storage manager is the component of a database that provides the interface between the low-level data stored in the database and the application programs and queries submitted to the system

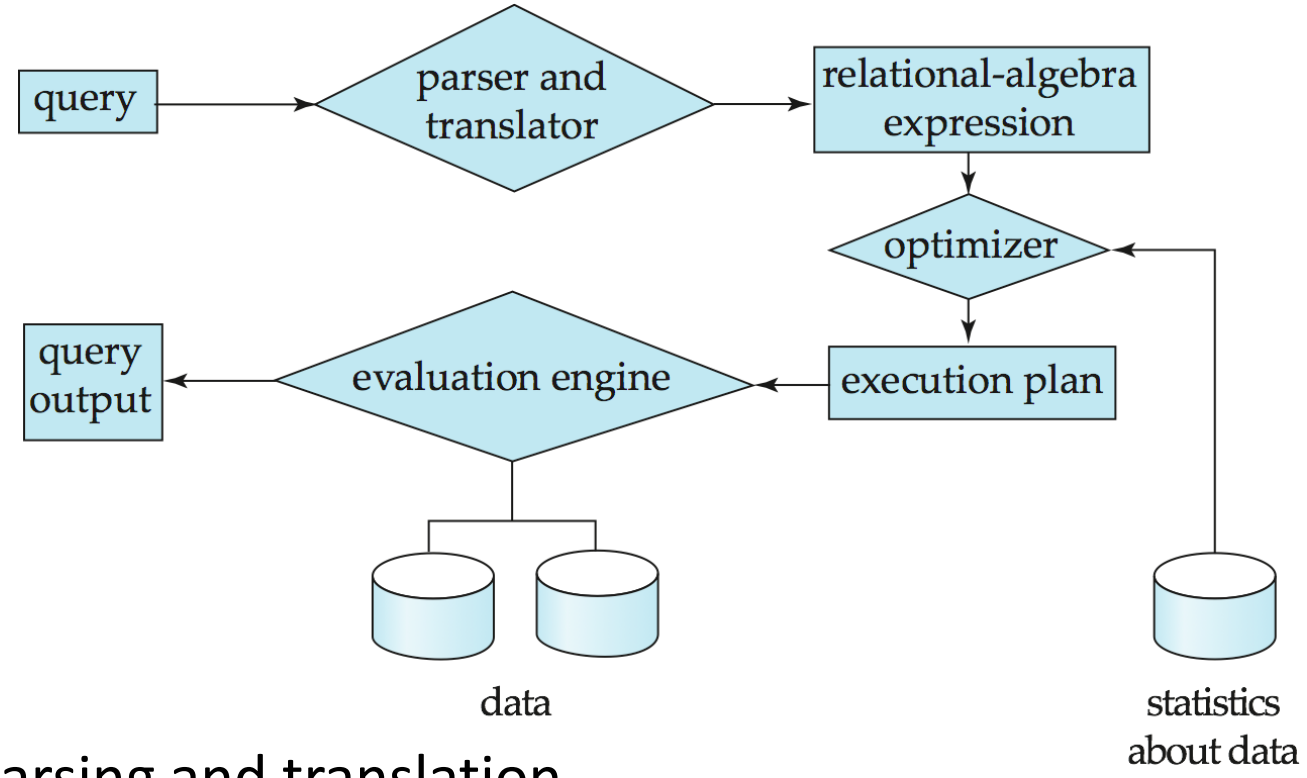


  - Authorization and integrity manager (impose integrity constraints)
  - Transaction manager (to ensure system consistency)
  - File manager (manages the allocation of space on disk storage)
  - Buffer manager (responsible for fetching data from disk storage to memory)

Data Storage and Querying

- The query processor components include:
  - DDL interpreter, which interprets DDL statements and records the definitions in the data dictionary.
  - DML compiler, which translates DML statements in a query language into an evaluation plan consisting of low-level instructions that the query evaluation engine understands.
  - Query evaluation engine, which executes low-level instructions generated by the DML compiler.
- The query processor helps the database system to simplify and facilitate access to data, allowing database users to obtain good performance while being able to work at the view level without understanding the physical-level details

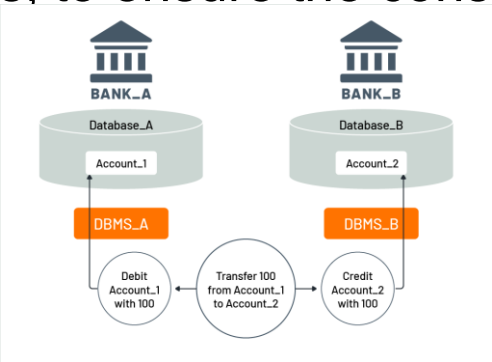
Query Processing



1. Parsing and translation
2. Optimization
3. Evaluation

Transaction Management

- What if the system fails?
- What if more than one user is concurrently updating the same data?
- A transaction is a collection of operations that performs a single logical function in a database application
- Transaction-management component ensures that the database remains in a consistent (correct) state despite system failures (e.g., power failures and operating system crashes) and transaction failures.
- Concurrency-control manager controls the interaction among the concurrent transactions, to ensure the consistency of the database.

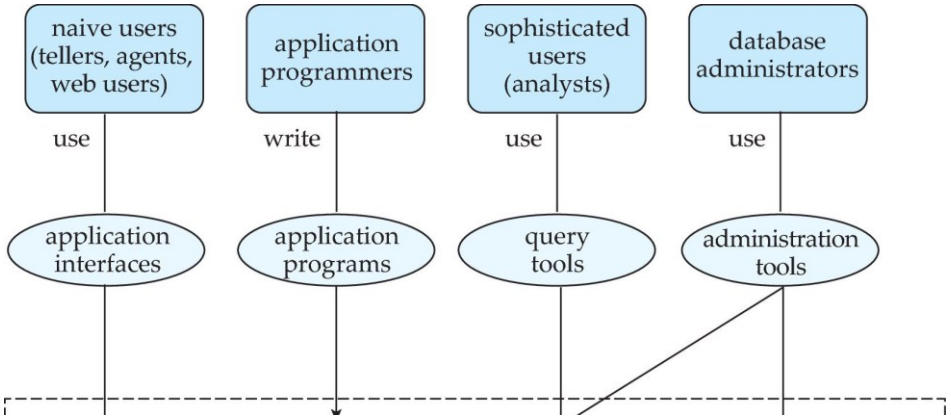


Source: Google Image

Database Users and Administrators

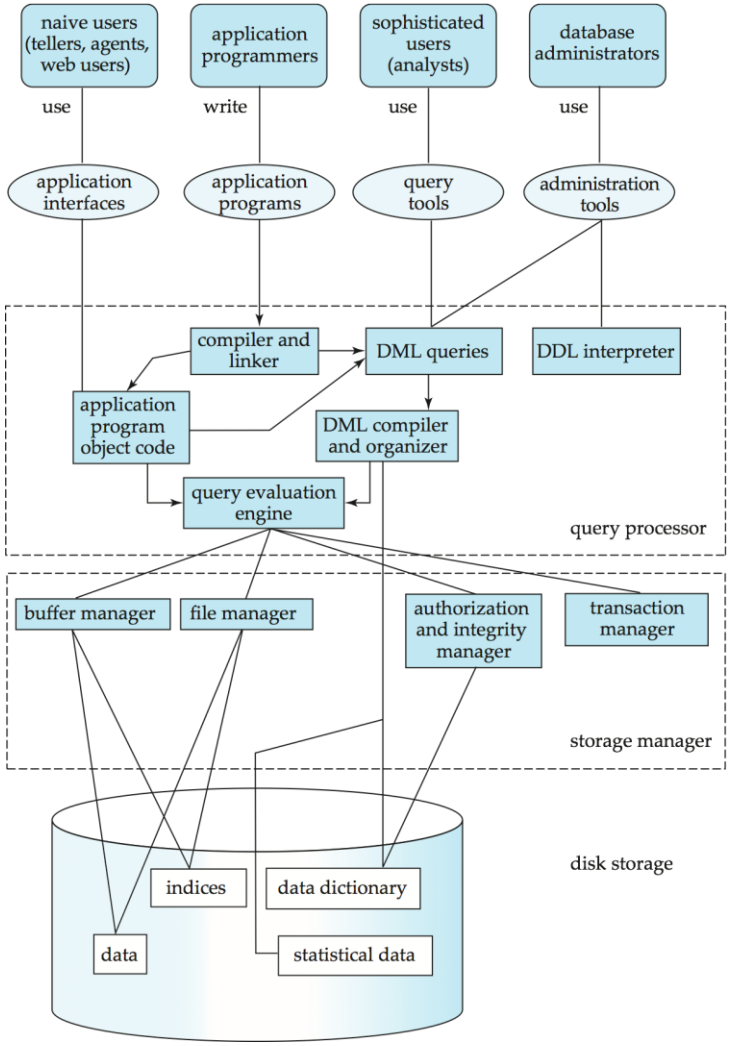
Who Uses the Database Systems

- Database users
  - Naive users
  - Application programmers
  - Sophisticated users



- Database administrators
  - Schema definition
  - Storage structure and access-method definition
  - Schema and physical-organization modification
  - Granting or authorization for data access
  - Routine maintenance (e.g., back up database, check disk space)

Database System Internals



Source: http://www.db-book.com/

Summary



- Database management systems (DBMS)
  - A collection of interrelated data and a set of programs to access those data
- The three levels of abstraction provided by the DBMS
  - Physical level
  - Logical level
  - View level
- Database languages
  - DDL
  - DML

Summary (Cont.)



- Relational model and database
  - A relational model uses a collection of tables to represent both the data and the relationships among those data.
  - Relational model is the underlying data model for relational databases

- Who uses databases?
  - Naive users
  - Application programmers
  - Sophisticated users
  - Database administrators



Thank you!

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Source: Google Image